Week-8

1.1 Filtering data: WHERE, AND, OR

The **WHERE** clause is used to filter records. The WHERE clause is used to extract only those records that fulfill a specified condition**.** The syntax of SELECT after incorporating WHERE clause.

```
SELECT <Attribute List>
FROM <Table List>
WHERE <Condition>;
```

Where <Condition> is a conditional expression that identifies the tuples(rows) to be retrieved by the Query.

Example 1: This query involves only the EMPLOYEE relation listed in the FROM clause. The query selects the individual EMPLOYEE tuples that satisfy the condition of the WHERE clause, then projects the result on the Bdate and Address attributes listed in the SELECT clause.

```
SELECT Bdate,Address
FROM Employee
WHERE ssn=121;
```

Operators in The WHERE Clause are

Operator	**Description**
=	Equal
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal
<>	Not equal
BETWEEN	Between a certain range
LIKE	Search for a pattern
IN	To specify multiple possible values for a column

AND

The **AND** operator allows the existence of multiple conditions in a SQL statement's WHERE clause. While using AND operator, complete condition will be assumed true when all the conditions are true.

Consider the table EMPLOYEE having records as follows –

EMPLOYEE

```
 id | name   | age | address    | salary
----+--------+-----+-----------+--------
  1 | Paul   |  32 | California|  20000
  2 | Allen  |  25 | Texas      |  15000
  3 | Teddy  |  23 | Norway     |  20000
  4 | Mark   |  25 | Rich-Mond  |  65000
  5 | David  |  27 | Texas      |  85000
  6 | Kim    |  22 | South-Hall|  45000
  7 | James  |  24 | Houston    |  10000
```

Query 0. Retrieve the details of all employees where AGE is greater than or equal to 25 AND salary is greater than or equal to 65000.

```
SELECT * FROM EMPLOYEE WHERE AGE >= 25 AND SALARY >= 65000;
```

Only those tuples that satisfy the condition—that is, those tuples for which the condition evaluates to TRUE after substituting their corresponding attribute values—are selected

Output

```
 id | name   | age | address     | salary
----+--------+-----+------------+--------
  4 | Mark   |  25 | Rich-Mond   |  65000
  5 | David  |  27 | Texas       |  85000
(2 rows)
```

OR

The OR operator is also used to combine multiple conditions in a SQL statement's WHERE clause. While using OR operator, complete condition will be assumed true when at least any of the conditions is true. For example [condition1] OR [condition2] will be true if either condition1 or condition2 is true.

Syntax

The basic syntax of OR operator with WHERE clause is as follows –

```
SELECT column1, column2, columnN
FROM table_name
WHERE [condition1] OR [condition2]...OR [conditionN]
```

The following SELECT statement lists down all the records where AGE is greater than or equal to 25 **OR** salary is greater than or equal to 65000.00 −

```
SELECT * FROM EMPLOYEE WHERE AGE >= 25 OR SALARY >= 65000;
```

The above given SQL statement will produce the following result

Output

```
 id | name  | age | address     | salary
----+-------+-----+-------------+--------
  1 | Paul  |  32 | California |  20000
  2 | Allen |  25 | Texas      |  15000
  4 | Mark  |  25 | Rich-Mond  |  65000
  5 | David |  27 | Texas      |  85000
(4 rows)
```

1.2 **Row limiting clause**: This clause allows sql queries to limit the number of rows returned and to specify a starting row for the return set.

Syntax:

```
SELECT column_name(s)
FROM table_name
ORDER BY column_name(s)
FETCH FIRST number ROWS ONLY;
```

Example:

The following SQL statement selects the first three records from the "Customers" table

```
SELECT * FROM Customers
FETCH FIRST 3 ROWS ONLY;
```

1.3 Filtering data:IN, BETWEEN, LIKE

IN

IN operator in the WHERE clause to check if a value matches any value in a list of values.

The syntax of the IN operator is as follows:

```
SELECT * FROM TABLE_NAME WHERE value IN (value1,value2,...)
```

The IN operator returns true if the value matches any value in the list i.e., value1 , value2 ,.The list of values can be a list of literal values such as numbers, strings or a result of a SELECT statement like this:

```
SELECT * FROM TABLE_NAME WHERE value IN (SELECT column_name FROM table_name);
```

Example:

```
SELECT * FROM EMPLOYEE WHERE ID IN (1,2,3);
```

The above Query display output as follows.

```
 id | name   | age | address    | salary
----+--------+-----+------------+--------
  1 | Paul   |  32 | California|  20000
  2 | Allen  |  25 | Texas      |  15000
  3 | Teddy  |  23 | Norway     |  20000
```

BETWEEN

We use the BETWEEN operator to match a value against a range of values. The following illustrates the syntax of the BETWEEN operator:

```
SELECT * FROM table_name WHERE  value BETWEEN low AND high;
```

If the value is greater than or equal to the low value and less than or equal to the high value, the expression returns true, otherwise, it returns false.

You can rewrite the BETWEEN operator by using the greater than or equal (>=) or less than or equal (<=) operators like this:

```
SELECT * FROM table_name WHERE  value >= low and value <= high;
```

Example:

```
SELECT * FROM EMPLOYEE WHERE salary BETWEEN 20000 and 75000;
```

The above Query display output as follows

```
 id | name   | age | address    | salary
----+--------+-----+------------+--------
  1 | Paul   |  32 | California|  20000
  3 | Teddy  |  23 | Norway     |  20000
  4 | Mark   |  25 | Rich-Mond  |  65000
  6 | Kim    |  22 | South-Hall|  45000
```

LIKE

The **LIKE** operator is used to match text values against a pattern using wildcards. If the search expression can be matched to the pattern expression, the LIKE operator will return true, which is **1**.

There are two wildcards used in conjunction with the LIKE operator −

- The percent sign (%)
- The underscore (_)

The percent sign represents zero, one, or multiple numbers or characters. The underscore represents a single number or character. These symbols can be used in combinations.

If either of these two signs is not used in conjunction with the LIKE clause, then the LIKE acts like the equals operator.

Syntax

The basic syntax of % and _ is as follows −

```
SELECT FROM table_name WHERE column LIKE '%XXXX%
```

And

```
SELECT FROM table_name WHERE column LIKE '_XXXX
```

You can combine any number of conditions using AND or OR operators. Here XXXX could be any numeric or string value.

For Example 1 :

```
SELECT * FROM EMPLOYEE WHERE AGE::text LIKE '2%'
```

This would produce the following result

```
 id | name  | age | address     | salary
----+-------+-----+-------------+--------
  2 | Allen |  25 | Texas       |  15000
  3 | Teddy |  23 | Norway      |  20000
  4 | Mark  |  25 | Rich-Mond   |  65000
  5 | David |  27 | Texas       |  85000
  6 | Kim   |  22 | South-Hall  |  45000
  7 | James |  24 | Houston     |  10000
  8 | Paul  |  24 | Houston     |  20000
(7 rows)
```

Which would display all the records from EMPLOYEE table where AGE starts with 2?

Example 2:

```
SELECT * FROM EMPLOYEE WHERE ADDRESS  LIKE '%-%';
```

This would produce the following result

```
 id | name | age |                 address                   | salary
----+------+-----+-------------------------------------------+--------
  4 | Mark |  25 | Rich-Mond                                 |  65000
  6 | Kim  |  22 | South-Hall                                |  45000
(2 rows)
```